

Life in oceans

In the billion or more years since the first animal species evolved, competition for survival in the sea has become ever more intense. Today, the oceans are home to the largest predators on the planet, as well as to vast numbers of microscopic animals that drift by unseen. As on land, ocean wildlife is affected by local conditions, the most important being the supply of food. For many, survival also depends on being able to defend themselves against attack. In some regions, life is thinly spread, but in others, animals are found in greater numbers than they are anywhere else on earth.

Feeding at sea

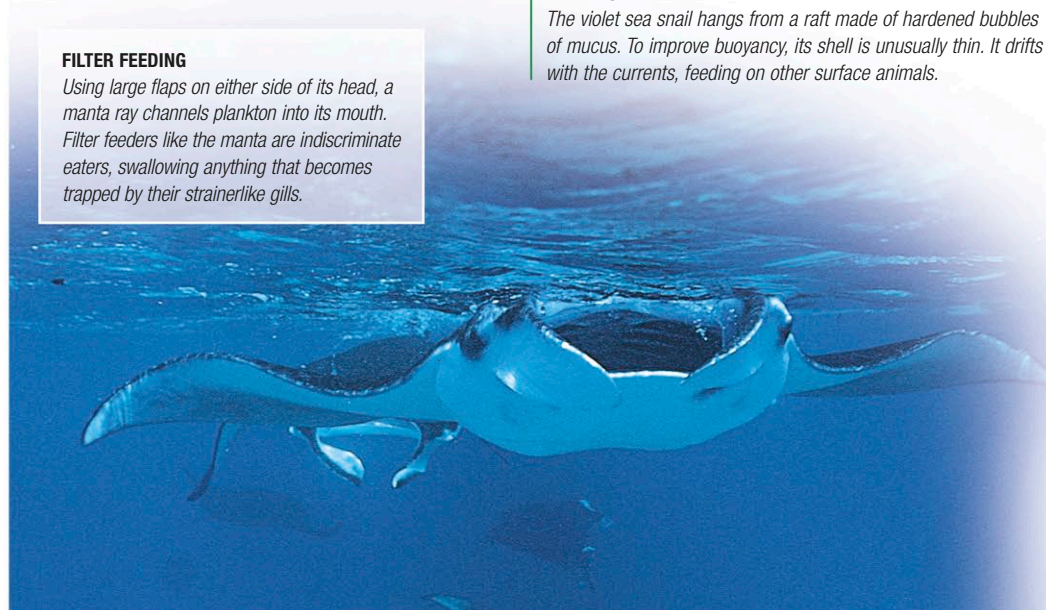
The oceans are so huge that, although they contain plenty of nutritious food, marine animals face a challenge in finding enough to eat without expending too much energy in the process.

Some concentrate on large prey. The sperm whale, for example, hunts giant squid at depths of over 3,300 ft (1,000 m), although most pursuit hunters search for food near the surface rather than in the depths. Other large sea animals eat smaller fare, scooping it up in huge amounts, often straining it with their gills. This technique, known as filter feeding, is used by baleen whales and some of the largest sharks and rays. Most filter feeders live on plankton, which is so abundant that it allows them to reach a gigantic size. Drifting animals also feed on plankton, although on a much smaller scale. Comb jellies or sea gooseberries, for example, haul it in with sticky cells that work like fishing nets.

On the seabed, animal life depends almost entirely on the dead organic matter that drifts steadily down from above. Brittlestars are typical of these scavengers, collecting food particles with their arms. However, there are also predators—bizarre fish, for example—that hunt in the total darkness, on or near the seabed. Animal life at the bottom of the sea can be sparse, so these predators cannot afford to miss any opportunity to feed. Many

FILTER FEEDING

Using large flaps on either side of its head, a manta ray channels plankton into its mouth. Filter feeders like the manta are indiscriminate eaters, swallowing anything that becomes trapped by their strainerlike gills.



of them therefore have gigantic mouths and elastic stomachs that enable them to swallow prey that is almost as large as themselves.

Controlling buoyancy

Very few marine animals—with the exception of deep-diving mammals—are found at all levels in the sea. Instead, most are adapted for life at a particular depth, and have buoyancy devices that help to keep them there. Surface drifters, such as the violet sea snail, have simple floats: filled with material that is lighter than water, they ensure that the animal stays at the surface, even in a heavy swell.

For animals that spend their lives fully immersed, remaining at one level requires more complex apparatus. They have to be neutrally buoyant at their optimal depth and yet able to rise or sink as the need arises. To do this, some use adjustable buoyancy aids hidden inside their bodies. Bony fish, for



ANIMALS ADRIFT

The violet sea snail hangs from a raft made of hardened bubbles of mucus. To improve buoyancy, its shell is unusually thin. It drifts with the currents, feeding on other surface animals.

BIOLUMINESCENCE

A variety of marine animals produce light—some to maintain contact with their own kind, others to lure prey. In some species, notably planktonic invertebrates, the value of light is less easy to explain. Bioluminescence is most common in bathypelagic species (those that live in very deep, open water). Typically, the light is produced by skin organs called photophores. Light production is not always confined to the animal itself: some fish can eject luminous clouds to distract predators while they make an escape.



LIGHTS IN THE DARK

Bristlemouths have photophores that emit a yellowish green light. As in most luminous fish, the photophores are arranged in characteristic patterns that enable members of a species to recognize each other in the dark. Photophores may shine for extended periods or they may be flashed on and off.

example, have a gas-filled chamber, called the swim bladder, just below the backbone. If the fish needs to sink, it removes some of the gas from the swim bladder by pumping it into the bloodstream; if it needs to rise, it pumps it back into the bladder. Cartilaginous fish, such as sharks, do not have swim bladders; instead, they rely on their large, oily liver to keep them afloat. Many sharks are actually slightly heavier than sea water: swimming provides the lift that allows them to control their depth.

Avoiding predators

In the open sea, there is nowhere to hide, which leaves animals highly vulnerable to predators. To survive, some rely on camouflage or disguise; others behave in ways that make them difficult to attack.

For slow-moving invertebrates, such as those that make up plankton, one of the most effective disguises is transparency. Planktonic animals are often as clear as glass, which makes them difficult to see, even at close quarters. Most of these animals are only a few millimeters long, although some tunicates form translucent, tube-shaped colonies that can be over 9¾ ft (3 m) in length.

Some fish are transparent when they are very small, but then use camouflage of a different kind as they get older. Almost all species that live in brightly lit, open water have dark backs but much paler undersides. This pattern, known as countershading, protects fish in 2 ways: it hides them from predators deeper down by disguising their silhouette against the